

DYN MODE

Standard dynamic balance — steel wheels, spare tires, trailers

CARD 1 OF 6

CAE WHEEL BALANCER

WHEN TO USE

Steel wheels · Trailer wheels · Old alloys where clips won't show · Spare tires · NOT for premium alloys — clip will chip the paint. Use ALU-2 or ALU-S1 instead.

SETUP

- 1 Mount the wheel.** Clean rim, remove old weights. Standard bore: cone small end facing the machine. Recessed bore: cone large end facing the machine. Tighten the quick-release nut firmly.
- 2 Power on.** Machine boots to DYN mode by default.

MEASURE WHEEL — A, B, D

- 1 a (distance):** Pull the inner gauge arm out and touch the inner rim edge. Hold 4 seconds. Value auto-saves, then return gauge to zero position.
- 2 d (diameter):** Captured automatically at the same time as 'a' — just read off the tire sidewall to confirm.
- 3 b (width):** Use the width gauge on top of the guard, or read the stamp on the rim. The sonar laser on the lid will auto-read when the hood closes.

SPIN AND APPLY WEIGHTS

- 1 Close the guard.** Press **START**. Spin takes ~8 seconds. Machine shows inner and outer imbalance in grams.
- 2 Rotate the wheel** by hand until the RIGHT (outer) LED fully lights up — this is the heavy spot. Clip the weight on the OUTER rim flange at the **12 o'clock** position.
- 3 Rotate again** until the LEFT (inner) LED fully lights up. Clip the weight on the INNER rim flange at **12 o'clock**.
- 4 Re-spin to verify.** Close the guard, press **START**. Display should read **0 / 0**. If not, check that weights didn't shift — re-seat or adjust.

QUICK TIP

If the imbalance is over 3 oz on a passenger wheel, don't just add weight — investigate first. Check for old weights missed, clean the bead seat, try match-mounting (see OPT card).

WHICH MODE?

Decision guide — pick the right balance mode for each wheel

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CAE WHEEL BALANCER

Look at the wheel. Answer the questions in order. Use the first mode that fits.

WHEEL TYPE / SITUATION	MODE	WEIGHT PLACEMENT
Steel wheel	DYN	Clip inside + clip outside (12 o'clock both)
Old/budget alloy, weights can be visible	DYN	Clip inside + clip outside (12 o'clock both)
Alloy — weights must be partially hidden	ALU-2	Clip inside + stick-on outer shoulder
Premium alloy — no clips anywhere	ALU-S1	Stick-on both sides, gauge-arm measured
Premium alloy — hide behind spokes	ALU-S1 + SPLIT	Two stick-ons behind adjacent spokes
Motorcycle / scooter wheel	Moto-1 or Moto-2	Single or paired adhesive

HOW TO CHANGE MODE

Press the ◀ (left) key repeatedly — the mode icon at the top of the screen cycles through DYN → ALU-1 → ALU-2 → ALU-3 → ALU-4 → ALU-5 → ALU-S1 → ALU-S2 → Moto-1 → Moto-2. Stop on the one you want — no confirm button needed, the mode is active as soon as its icon is lit.

WHY MODE MATTERS

- **Clips on painted alloys chip the finish.** One hammer swing = \$50+ wheel refinishing.
- **Clips can fly off curved-lip alloys at highway speed.** They need a proper flange to seat.
- **Stick-ons need a clean, flat surface.** Wipe the inside barrel with brake cleaner before applying.
- **Wrong mode = wrong weight position.** The machine calculates assuming specific weight planes — mode tells it which planes.

ALU-S1 + SPLIT WEIGHT

Premium alloys — weights hidden behind spokes

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CAE WHEEL BALANCER

WHEN TO USE

- Gloss black, polished, or chrome alloys
- Wheels where the customer will see any weight
- High-performance / show / EV wheels with tight spoke profiles

SETUP — ALU-S1 MODE

- 1 Clean the inside barrel first.** Wipe the areas where stick-ons will go with brake cleaner or isopropyl alcohol. Let flash off. Tape won't hold long-term on oily aluminum.
- 2** Mount the wheel. Press ◀ until **ALU-S1** icon lights up.
- 3 Set al (inner position):** Pull gauge arm out, touch the inner stick-on target location, hold 4 seconds.
- 4 Set aE (outer position):** Pull gauge arm out further, touch the outer stick-on target, hold 4 seconds.
- 5 Set dl (inner rim diameter):** Auto-captured with al. Confirm reading matches.
- 6** Close guard, press **START**. Wait for spin to complete.

SPLIT WEIGHT (HIDE BEHIND SPOKES)

After the ALU-S1 spin finishes — use this to divide the outer weight between two spokes so nothing is visible from outside:

- 1** Press ► → select the spoke-split icon → press ▲.
- 2** Use ◀ ► to input the number of spokes on the wheel. Press **START**.
- 3** Rotate the wheel by hand until the FIRST spoke is at 12 o'clock. Press **START**.
- 4** Rotate the wheel until the NEXT spoke is at 12 o'clock. Press **START**.
- 5** Machine now shows the outer weight split into SP1 and SP2 values — two smaller stick-ons, each placed directly behind a spoke.
- 6** Press ▼ until each indicator (SP1, SP2, inner) lights fully, apply weight at 6 o'clock each time. Re-spin to verify 0 / 0.

BEFORE YOU STICK

Verify the stick-on will clear the brake caliper — measure clearance on a test wheel once per vehicle model. Low-profile stick-ons are available if the standard strip is too tall. Keep a box in stock.

CALIBRATION

When to run each calibration and how

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CAE WHEEL BALANCER

From the home screen, press ► then select the calibration icon, then press ▲ to enter.

CALIBRATION	WHEN	REQUIRES
Self-calibration (100 g)	Monthly · after moving machine · if readings seem off	14-18" wheel, 100 g clip weight
Rim distance gauge	If 'a' reading is wrong · after bumping the arm	Nothing mounted
Diameter gauge	If 'd' reading doesn't match the tire sidewall	Any tire mounted
Width gauge (sonar/radar)	If 'b' auto-reading is off · after cleaning the laser	Known-width tire for compensation
Laser calibration	If the weight placement laser is off-centre	Nothing mounted
Self-diagnosis	When troubleshooting — checks all sensors	Nothing mounted

SELF-CALIBRATION PROCEDURE (100 G)

- 1 Mount a medium wheel (14"–18") that accepts a clip weight. Set **a**, **b**, **d** values normally. Close guard. Press **START**.
- 2 After the spin, clip a **100 g weight** on the OUTSIDE rim flange at **12 o'clock**. Close guard. Press **START**.
- 3 Move the same 100 g weight to the INSIDE rim flange at **12 o'clock**. Close guard. Press **START**.
- 4 Screen shows **OK** when finished. Machine is now calibrated.

CALIBRATION WEIGHT

Use the exact 100 g weight that came with the machine, or a verified 100 g weight.
An inaccurate reference weight will calibrate the machine INCORRECTLY — every balance after will be wrong.

ERROR CODES

What each ERR code means and how to fix it

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CODE	MEANING	FIX
ERR:1	No spin, or shaft spinning incorrectly	Check/replace power board · Check position-pickup board · Re-align pickup bracket
ERR:2	No wheel attached, or wheel not locked tight enough	Tighten quick-release nut firmly · If still errors, check position-pickup board
ERR:3	Beyond work scope — tire pressure low or wheel distorted	Inflate tire to proper PSI · Inspect for bent rim · Replace if damaged
ERR:4	Position-pickup or computer board fault	Check/replace position-pickup board · Check/replace computer board
ERR:5	Micro-switch (guard safety) problem	Check/replace the guard micro-switch · Check computer board
ERR:6	Power board or computer board fault	Check/replace power board · Check/replace computer board
ERR:7	Program lost / memory corruption	Run self-calibration (100 g procedure) · If persists, computer board fault
ERR:8	100 g weight was NOT added during self-calibration	Re-run self-calibration with the 100 g weight clipped as instructed
ERR:9	Data protection / locked	Contact supplier to unlock · Update firmware/data

BEFORE YOU CALL SERVICE — CHECK THESE FIRST

- Guard fully closed? ERR:5 and no-spin issues often trace to a loose or misaligned safety hood.
- Wheel mounted tight? A loose quick-release nut throws weird errors and bad reads.
- Floor level and solid? Vibration from nearby equipment or a wobbly floor kills accuracy.
- Power clean? Extension cords and shared circuits can trip the power board. Run a dedicated 110V circuit.
- Run self-diagnosis mode (■ → stethoscope icon → ▲) to see raw sensor values — tells you which board is faulty.

SUPPLIER CONTACT

Canadian Auto Equipment · 1-888-269-3346 · customercare@canadianautoequipment.ca
Model: CAE-3446WB · Warranty: 2 yr structural, 1 yr non-structural, 90 days wear

OPT FUNCTION

High imbalance fix — guided match-mount routine

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CAE WHEEL BALANCER

WHEN TO USE

Use OPT whenever a wheel calls for more than **3 oz (85 g)** of weight on a passenger tire. OPT is the balancer's built-in match-mount routine — it tells you exactly where to rotate the tire on the rim so the heavy spots cancel instead of adding up. Since this balancer has no road-force, OPT is your best tool for catching bad balance jobs before the customer feels a vibration at 100 km/h.

PROCEDURE

- 1 Mount the wheel. Enter **a, b, d** as normal. Press ►, select the **OPT** icon, press ▲.
- 2 Close guard. Press **START**. First spin measures the baseline imbalance.
- 3 **Break the tire off the rim and rotate it 180°** using your tire changer. Re-seat, re-inflate. Press ▲.
- 4 Close guard. Press **START** for the second spin.
- 5 Press ▼ to rotate the wheel until **four indicators light up** (two on each side). Mark the tire with chalk at 12 o'clock — this is position **C**.
- 6 Press ▼ again until only **two indicators light up** (one on each side). Mark the rim at 12 o'clock with chalk — this is position **D**.
- 7 Break the tire off again. Rotate it on the rim so that the **C mark lines up with the D mark**. Re-seat, re-inflate.
- 8 Close guard. Press **START** for the final spin. Imbalance should now be much lower — apply weights normally.

WHY IT WORKS

Every tire has a heavy spot. Every wheel has a heavy spot. Mount them randomly and the heavy spots can stack up, calling for 4+ oz of weight. OPT measures both and tells you where to rotate the tire so the heavy spots sit opposite each other — they partly cancel, dropping the required weight to a normal level.

CUSTOMER TIP

If OPT drops the weight from 4.5 oz to under 2 oz — tell the customer. It's the kind of detail that separates a \$30 balance from a \$50 premium balance, and builds repeat business.